

SAMBHAV KUMAR MOHANTY

AI/LLM Application Engineer | Backend & GenAI Systems

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PROFESSIONAL SUMMARY

Software engineer with strong Java/Spring Boot backend foundations actively transitioning into AI/LLM application development. Experienced in building production-grade REST APIs, integrating LangChain/LangGraph pipelines, and deploying RAG-based systems with vector databases. Combines solid software engineering discipline with emerging GenAI expertise — uniquely positioned to bridge traditional backend systems and modern AI-powered applications. Seeking roles in AI/LLM engineering, GenAI product development, or AI-augmented backend systems.

TECHNICAL SKILLS

AI/ML & GenAI: LangChain, LangGraph, RAG Architecture, Vector Embeddings, Prompt Engineering, HuggingFace, NLTK, Scikit-learn, Pandas, NumPy, OpenCV, YOLOv8, TensorFlow

LLMs & Models: LLaMA 3.3 (70B), Gemini 2.0 Flash, DeepSeek R1, GPT-4, Claude, LoRA/QLoRA fine-tuning concepts

Vector Databases: ChromaDB, Pinecone, FAISS, Qdrant

Backend & APIs: Java, Spring Boot, Spring MVC, FastAPI, Flask, Django, RESTful APIs, JWT Authentication, SQLAlchemy, WebSockets, Celery

Databases: PostgreSQL, MySQL, SQLite, Redis, NoSQL Basics

DevOps & Cloud: Docker, Git/GitHub, CI/CD, Vercel, Render, Linux, AWS Basics (EC2/RDS), Bash Scripting

Languages: Python, Java, SQL, JavaScript, HTML/CSS

Tools & IDEs: VS Code, IntelliJ IDEA, Postman, Jupyter Notebook, Maven, Swagger

EXPERIENCE

Programmer Analyst Trainee | [Cognizant](#) May 2025 – Present • Chennai, India

- Led a 5-member developer team to deliver a Spring Boot e-commerce platform, improving code quality and delivery efficiency by 30%.
- Engineered optimized RESTful APIs improving backend response times by 25% and strengthening database interactions across core service modules.
- Accelerated backend skill ramp-up by transitioning from Python to Java within 3 weeks, enabling full contribution to production-grade sprint deliverables.
- Integrated Spring Boot microservice components reducing processing overhead by 15% across key service endpoints.

CV Tech Intern | [HighRadius](#) Jun 2024 – Jan 2025 • Bhubaneswar, India

- Analyzed and optimized MySQL datasets using SQLYOG, reducing data inconsistencies by 25% across Order-to-Cash (O2C) operations.
- Streamlined payment processing and remittance handling workflows, improving transactional accuracy by 20%.
- Diagnosed and resolved complex customer issues across ERP modules via Salesforce CRM, achieving 4.2/5 CSAT across 50+ client tickets and earning Top Performer recognition (Nov 2024).
- Gained end-to-end exposure to O2C cycle, Credit Risk Management, and Cash Application processes in a production ERP environment.

PROJECTS

Multi-Agent AI Research Assistant | [LangGraph](#), [LangChain](#), [RAG](#), [FastAPI](#), [ChromaDB](#), [Docker](#) 2025

- Architected a LangGraph node-based multi-agent system with approval workflows, preventing 95% unauthorized state transitions and ensuring reliable task orchestration.
- Built a RAG pipeline with ChromaDB vector store achieving 85% context retention across multi-turn conversations through advanced state management.
- Integrated real-time SQLite synchronization for persistent memory management, supporting 500+ document ingestion and retrieval workflows.
- Deployed system via FastAPI with Docker containerization, enabling scalable multi-agent query resolution on cloud infrastructure.

NLP-Powered Semantic Similarity Engine | Python, NLTK, Scikit-learn, Levenshtein Algorithm2025

- Developed a sentence similarity detection system achieving 78% accuracy using Levenshtein distance and fuzzy matching algorithms.
- Implemented preprocessing pipeline with stop word removal and stemming, improving classification accuracy by 15%.
- Processed 10,000+ sentence pairs for plagiarism detection and content deduplication, demonstrating production-scale NLP pipeline design.

Real-Time Face Recognition Attendance System | OpenCV, Python, Face Recognition API, SQLite2025

- Built a contactless attendance system achieving 95% recognition accuracy processing 30 FPS real-time video streams using OpenCV and Face Recognition API.
- Engineered enrollment and verification pipeline with SQLite logging, reducing manual attendance overhead by 80%.

Mockito E-Commerce Platform (Spring Boot) | Java, Spring Boot, REST APIs, JUnit, MockitoAug – Oct 2025

- Built full-stack Spring Boot e-commerce platform with REST APIs, JWT authentication, admin dashboards, and product management modules.
- Integrated Mockito-based unit test suites achieving 40% improvement in code reliability through structured test coverage and error handling.

EDUCATION

B.Tech – Computer Science & Engineering2021 – 2025

Institute of Technical Education & Research, SOA University, BhubaneswarGPA: 8.66 / 10

CERTIFICATIONS & ACHIEVEMENTS

Certifications: Microsoft Azure Fundamentals (AZ-900) • Machine Learning A-Z: Python & R (Udemy) • The Ultimate MySQL Bootcamp • Git Complete • HackerRank Verified: Python, Java, MySQL (Basic)

Problem Solving: 300+ coding challenges solved across LeetCode, HackerRank, and GeeksforGeeks

Open Source: Contributor to AI/ML projects on GitHub with 50+ stars across repositories

Achievements: Top 10 – Cognizant Spring Boot Training Leaderboard • Top Performer – HighRadius (Nov 2024) • Top 15 Finalist – Smart India Hackathon 2024